Experiment 1

Getting started with Basics of Network configurations files and Networking Commands in Linux.

The important network configuration files in Linux operating systems are

1. /etc/hosts
The **/etc/hosts** file contains the Internet Protocol (IP) host names and addresses for the local host and other hosts in the Internet network. This file is used to resolve a name into an address (that is, to translate a host name into its Internet address).
127.0.1.1 anil-300E4Z-300E5Z-300E7Z

2. /etc/resolv.conf
This configuration file contains the IP addresses of DNS servers and the search domain.
A sample file is shown
DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
nameserver 127.0.1.1

3. /etc/sysconfig/network
This configuration file specifies routing and host information for all network interfaces. It contains directives that are global specific. For example if NETWORKING=yes, then /etc/init.d/network activates network devices.

4. /etc/nsswitch.conf
This file includes database search entries. The directive specifies which database is to be searched first.

The important Linux networking commands are

1. ifconfig

The ifconfig command is used to configure and display network interfaces on a Unix-like operating system. It's commonly used to check network settings, such as IP addresses, network masks, and other interface information.

2. netstat
The netstat command is a network utility used to display network connections, routing tables, interface statistics, and other network-related information on a system. It is commonly used to troubleshoot networking issues and to gather information about the network configuration and activity on a system.

Proto	Recv-Q	Send-Q	Local Address	Foreign Address	State
tcp	0	0	0.0.0.0:22	0.0.0.0:	LISTEN
tcp	0	0	192.168.1.100:22	192.168.1.101:22	ESTABLISHED

3. ping
This is the most commonly used command for checking connectivity.

ping www.google.com
PING www.google.com (172.217.163.36) 56(84) bytes of data.
64 bytes from maa05s01-in-f4.1e100.net (172.217.163.36): icmp_seq=1 ttl=53 time=51.4 ms
--- www.google.com ping statistics ---

1 packet transmitted, 1 received, 0% packet loss, time 7004ms
rtt min/avg/max/mdev = 48.533/56.804/68.266/6.030 ms

4. T**raceroute**

It is used to get the route of a packet. In other words, the traceroute command is used to determine the path along which a packet travels. It also returns the number of hops taken by the packet to reach the destination.

Eg, traceroute google.com
traceroute to example.com (93.184.216.34), 30 hops max, 60 byte packets
1 192.168.1.1 (192.168.1.1) 0.452 ms 0.420 ms 0.412 ms
2 10.10.10.1 (10.10.10.1) 2.134 ms 2.108 ms 2.091 ms

5. arp

When a device wants to communicate with another device on the local network, it needs to know the MAC address associated with the IP address of the destination device. If the device does not already have the MAC address cached, it sends out an ARP request

eg; arp

Address	HWtype	HWaddress	Flags Mask	Iface
192.168.11.225	ether	6c:3c:7c:c7:53:e0	C	eno1
_gateway	ether	8c:c7:c3:40:84:43	C	eno1